B3 past final worksheet

1. (Fall 2023, #6) Consider the function

$$f(x) = \frac{x-18}{\sqrt{x^2+36}}$$

(a) Write the domain of f and find all its vertical asymptotes, if any:

(b) Find all horizontal asymptotes (if any) of f and justify by computing appropriate limits. *Hint:* $\sqrt{x^2} = |x|$ and there are two horizontal asymptotes.

(c) It can be shown that

$$f'(x) = \frac{18x+36}{(x^2+36)^{3/2}}$$

Find the intervals on which f is increasing/decreasing. Give your answer in interval notation.

(d) Find the x-coordinates of all local extrema of f and classify them as local max/min.

(e) It can be shown that

$$f''(x) = -\frac{36(x-3)(x+6)}{(x^2+36)^{5/2}}$$

Find the intervals on which f is concave up and concave down. Give your answer in interval notation.

(f) Find the x-coordinates of all inflection points of f.

2. (Spring 2023, #6) Consider the function $f(x) = \dfrac{e^x}{x}$. The first and second derivatives of $f(x)$ are provided below (you don't need to verify these derivatives).

$$f'(x) = \frac{e^x(x-1)}{x^2} \qquad\qquad f''(x) = \frac{e^x(x^2-2x+2)}{x^3}$$

(a) Identify any vertical asymptotes of $f(x)$, and use appropriate limits to determine the behavior of $f(x)$ on both sides of each asymptote you identify.

(b) Find the x-coordinates of all critical points of $f(x)$, determine the intervals of increase and decrease, and classify the critical points as local maxima, local minima, or neither.

(This problem continues on the next page.)

To assist your work on this page, we reproduce $f(x)$, $f'(x)$ and $f''(x)$ here.

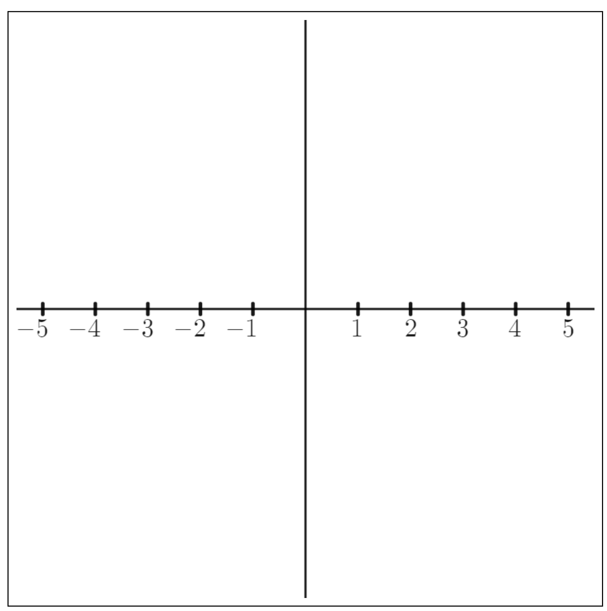
$$f(x) = \frac{e^x}{x} \qquad\qquad f'(x) = \frac{e^x(x-1)}{x^2} \qquad\qquad f''(x) = \frac{e^x(x^2-2x+2)}{x^3}$$

(c) Determine the intervals of positive concavity and negative concavity, and find the x-coordinates of any inflection points.

(d) It can be shown that

$$\lim_{x\to\infty} f(x) = \infty \qquad \text{and} \qquad \lim_{x\to-\infty} f(x) = 0$$

Use these facts, along with the information you found in parts (a)-(c) to sketch a graph of $f(x)$ on the axes provided to the right. Notice that there is no scale on the vertical axis, so your graph will not be graded by its vertical scale, but all features from parts (a)-(c) should be clearly visible in your sketch.



3. (Fall 2022, #6) Consider the function $f(x) = \dfrac{e^x}{x}$ and its first and second derivatives

$$f'(x) = \frac{(x-1)e^x}{x^2}, \qquad f''(x) = \frac{(x^2-2x+2)e^x}{x^3}$$

(a) Find the domain of f.

(b) Find all vertical asymptotes of f, or show that there are none. Determine the behavior of f on either side of any vertical asymptote that you find.

(c) Find all horizontal asymptotes of f, or show that there are non. Hint: $\lim\limits_{x\to\infty} \dfrac{e^x}{x} = +\infty$

(d) Find all zeros of f, or show that there are none.

(e) Find all critical points of f.

(f) Find all intervals where f is increasing and where f is decreasing, as well as any local min or max.

4. (Fall 2021, #7) Consider the following function and its first and second derivative:

$$f(x) = \frac{x^{\frac{4}{9}}}{e^{\frac{2x}{9}}} \qquad f'(x) = \frac{4-2x}{9x^{\frac{5}{9}}e^{\frac{2x}{9}}} \qquad f''(x) = \frac{4(x^2-4x-5)}{81x^{\frac{14}{9}}e^{\frac{2x}{9}}}$$

Additionally, $\lim_{x\to\infty} f(x) = 0$ and $\lim_{x\to-\infty} f(x) = \infty$.

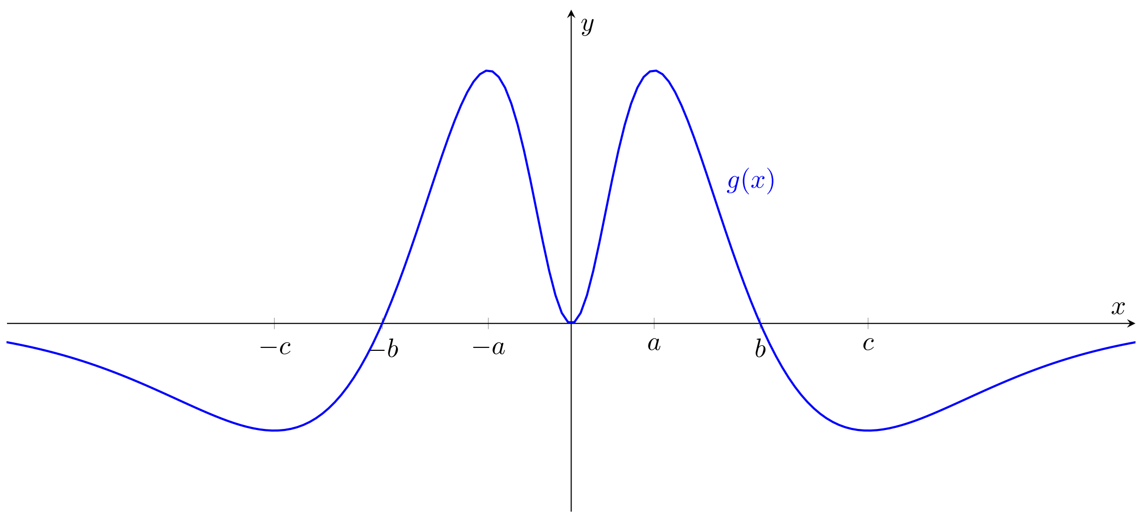
(a) Find all horizontal asymptotes of f.

(b) Compute two one-sided limits at each vertical asymptote of the derivative f'.

(c) Find all critical points of f and classify them using a sign table.

(d) Find the intervals where f is concave up or down and find any inflection points.

5. (Spring 2021, #6) The graph of a function $g(x)$ is shown below. As suggested by the graph, $g(0) = 0$ and $g'(0) = 0$. The other labeled points on the x-axis occur at the zeros and local extrema of $g(x)$.



Let $f(x)$ be a function whose derivative is $g(x)$, i.e. $f'(x) = g(x)$. Use the graph of $g(x)$ to answer the following questions about $f(x)$.

(a) On what intervals is $f(x)$ decreasing?

(b) What are the critical points of $f(x)$? Classify each as a local maximum, local minimum, or neither.

(c) At what points in the domain do the inflection points of $f(x)$ occur?

(d) On what intervals does $f(x)$ have positive concavity?

(e) Now suppose we also know the following information about $f(x)$:

- $f(0) = 0$
- $\lim_{x \to \infty} f(x) = 0$
- $\lim_{x \to -\infty} f(x) = 0$
- The absolute maximum value of $f(x)$ is d
- The absolute minimum value of $f(x)$ is $-d$

Use this information along with your answers to parts (a) through (d) to sketch a possible graph of $f(x)$ on the empty pair of axes below. Be careful to clearly draw the relevant features of the graph at the appropriate points of the domain. The graph of $g(x)$ is included at the bottom of the page for your reference.